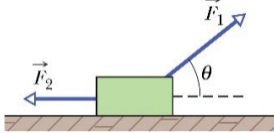


1-2 | Constant velocity and constant force

1. In the figure below, forces $\vec{F}_1$ and $\vec{F}_2$ are applied to a box as it slides at a constant velocity over a frictionless floor. We are to decrease the angle θ without changing the magnitude of $\vec{F}_1$. For constant velocity, should we increase, decrease, or maintain the magnitude of $\vec{F}_2$?



1. Increase the magnitude of F_2 .

Constant velocity means $a = 0$. Horizontal force balance gives $F_2 = F_1 \cos \theta$. When θ decreases, $\cos \theta$ increases, so F_2 must increase to balance the larger rightward component.

2. At time $t = 0$, constant force $\vec{F}$ begins to act on a rock moving through deep space in the $+x$ direction.
- a) For time $t > 0$, which are possible functions $x(t)$ for the rock's position:
- i. $x = 4t - 3$
 - ii. $x = -4t^2 + 6t - 3$
 - iii. $x = 4t^2 + 6t - 3$
- b) For which function is $\vec{F}$ directed opposite the rock's initial direction of motion?

2. (a) ii and iii. (b) ii.

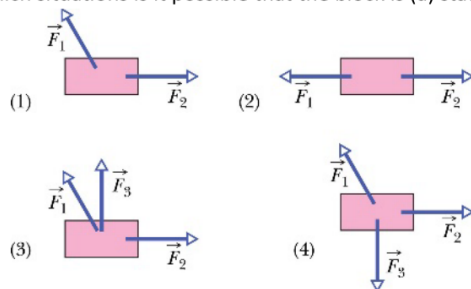
Function	$v = dx/dt$	$a = d^2x/dt^2$	Interpretation
i	4	0	Zero net force
ii	$-8t + 6$	-8	Constant force in -x
iii	$8t + 6$	+8	Constant force in +x

Both ii and iii have positive initial velocity, $v(0) = +6$, and constant nonzero acceleration. By $F = ma$, ii has force opposite the initial $+x$ motion.

Wording note: The intended reading is a nonzero force beginning to act. If zero force were allowed, i would also be a constant-force case.

3-4 | Balance forces and add components

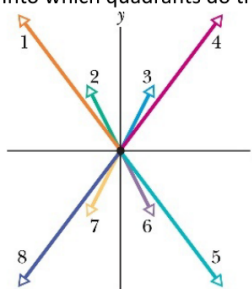
3. The figure below shows overhead views of four situations in which forces act on a block that lies on a frictionless floor. If the force magnitudes are chosen properly, in which situations is it possible that the block is (a) stationary and (b) moving with a constant velocity?



3. (a) Situations 2 and 4. (b) Situations 2 and 4.

Both remaining at rest and moving at constant velocity require zero net force. In 2, equal opposite forces cancel. In 4, $\vec{F}_2$ cancels the leftward part of $\vec{F}_1$ and $\vec{F}_3$ cancels its upward part. In 1 and 3, no downward force cancels the upward component.

4. Two horizontal forces, $\vec{F}_1 = (3 \text{ N})\hat{i} - (4 \text{ N})\hat{j}$ and $\vec{F}_2 = -(1 \text{ N})\hat{i} - (2 \text{ N})\hat{j}$ pull a banana split across a frictionless lunch counter. Without using a calculator, determine which of the vectors in the free-body diagram below best represent (a) $\vec{F}_1$ and (b) $\vec{F}_2$. What is the net force component along (c) the x axis and (d) the y axis? Into which quadrants do the (e) net-force vector and (f) acceleration vector point?



4. (a) 5; (b) 7; (c) +2 N; (d) -6 N; (e) IV; (f) IV.

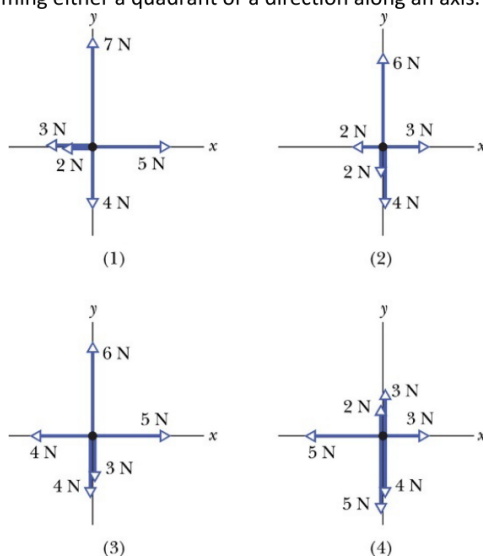
$\vec{F}_1 = (3, -4) \text{ N}$ is the longer down-right arrow (5). $\vec{F}_2 = (-1, -2) \text{ N}$ is the shorter down-left arrow (7).

$$\vec{F}_{\text{net},x} = 3 - 1 = +2 \text{ N}; \quad \vec{F}_{\text{net},y} = -4 - 2 = -6 \text{ N}.$$

The resultant points right and down, in quadrant IV. Acceleration points in the same direction as the net force.

5 | Acceleration from a force diagram

5. The figure below gives the free-body diagrams for four situations in which an object is pulled by several forces across a frictionless floor, as seen from overhead. In which situations does the object's acceleration $\vec{a}$ have (a) an x component and (b) a y component? (c) In each situation, give the direction of $\vec{a}$ by naming either a quadrant or a direction along an axis.



5. (a) 2, 3, 4. (b) 1, 3, 4.

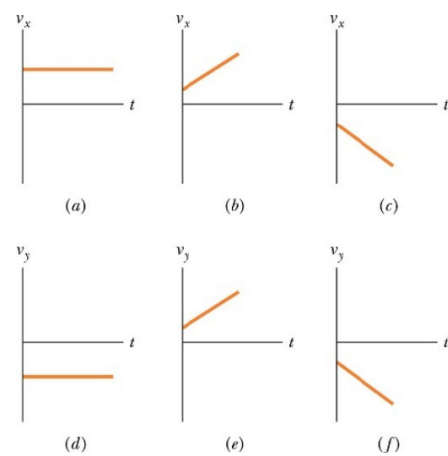
Situation	Net F_x (N)	Net F_y (N)	(c) Direction of $\vec{a}$
1	$5 - 3 - 2 = 0$	$7 - 4 = +3$	+y axis
2	$3 - 2 = +1$	$6 - 2 - 4 = 0$	+x axis
3	$5 - 4 = +1$	$6 - 3 - 4 = -1$	Quadrant IV
4	$3 - 5 = -2$	$2 + 3 - 4 - 5 = -4$	Quadrant III

Use $\vec{a}_x = \vec{F}_{\text{net},x}/m$ and $\vec{a}_y = \vec{F}_{\text{net},y}/m$. A component of acceleration is nonzero only when the net force component along that axis is nonzero.

Key idea: Individual forces can be present along an axis while their net force is zero. Add all arrows with signs before choosing the direction.

6-7 | Velocity slopes and connected blocks

6. The figure below gives three graphs of velocity component $v_x(t)$ and three graphs of velocity component $v_y(t)$. The graphs are not to scale. Which $v_x(t)$ graph and which $v_y(t)$ graph best correspond to each of the four situations in Question 5?

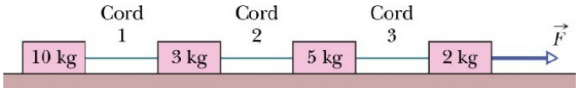


6. Match the signs of acceleration to the graph slopes.

Situation from Q5	Sign of a_x	Sign of a_y	v_x graph	v_y graph
1	0	+	a	e
2	+	0	b	d
3	+	-	b	f
4	-	-	c	f

Use $a_x = dv_x/dt$ and $a_y = dv_y/dt$. Forces determine slopes; initial velocities, and therefore graph heights, are not specified.

7. The figure below shows a train of four blocks being pulled across a frictionless floor by force $\vec{F}$. What total mass is accelerated to the right by (a) force $\vec{F}$, (b) cord 3, and (c) cord 1? (d) Rank the blocks according to their accelerations from greatest to least. (e) Rank the cords according to their tension from greatest to least.



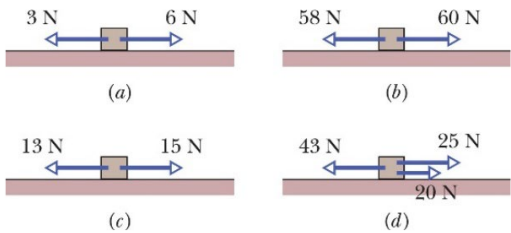
7. (a) 20 kg. (b) 18 kg. (c) 10 kg. (d) All equal. (e) $T_3 > T_2 > T_1$.

For taut, massless, inextensible cords, every block has $a = F/(20 \text{ kg})$. Each cord pulls the entire group to its left:

$T_1 = (10 \text{ kg})a = 0.50F$
 $T_2 = (13 \text{ kg})a = 0.65F$
 $T_3 = (18 \text{ kg})a = 0.90F$

8-9 | Net force and the normal force

8. The figure below shows the same object in four situations where horizontal forces are applied. Rank the situations according to the magnitude of the box’s acceleration from greatest to least.



8. $a > b = c = d$.

Situation	Net horizontal force (right positive)
a	$6 - 3 = +3 \text{ N}$
b	$60 - 58 = +2 \text{ N}$
c	$15 - 13 = +2 \text{ N}$
d	$25 + 20 - 43 = +2 \text{ N}$

The object has the same mass in every case, so $|a| = |F_{\text{net}}|/m$. Compare the net forces, not individual force magnitudes.

9. A vertical force $\vec{F}$ is applied to a block of mass m that lies on the floor. What happens to the magnitude of the normal force $\vec{F}_N$ on the block from the floor as magnitude F is increased from zero if force $\vec{F}$ is (a) downward and (b) upward?

9. (a) Increases: $N = mg + F$. (b) Decreases to zero.

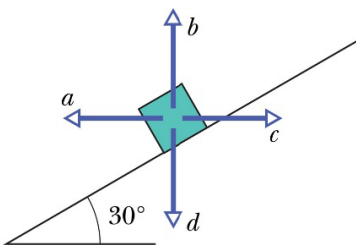
Downward applied force: While the block remains on the floor, $N - mg - F = 0$, so $N = mg + F$. The floor supports both weight and the extra downward push.

Upward applied force: While contact remains, $N + F - mg = 0$, so $N = mg - F$. At $F = mg$, $N = 0$. For $F > mg$, the block lifts off and N stays zero.

For an upward pull: $N = \max(mg - F, 0)$. The normal force cannot become negative; the floor cannot pull the block downward.

10 | Normal force on an incline

10. The figure below shows four choices for the direction of a force of magnitude F to be applied to a block on an inclined plane. The directions are either horizontal or vertical. (For choices a and b , the force is not enough to lift the block up off of the plane. Rank the choices according to the magnitude of the normal force on the block from the plane from greatest to least.



10. $d > c > a > b$.

Resolve all forces perpendicular to the 30° plane. The weight component into the surface is $mg \cos 30^\circ$. An applied component into the plane increases N ; an applied component away from the plane decreases N .

Choice	Applied direction	Normal force
a	Left	$N_a = mg \cos 30^\circ - F \sin 30^\circ$
b	Up	$N_b = mg \cos 30^\circ - F \cos 30^\circ$
c	Right	$N_c = mg \cos 30^\circ + F \sin 30^\circ$
d	Down	$N_d = mg \cos 30^\circ + F \cos 30^\circ$

Because $\cos 30^\circ = \sqrt{3}/2 > \sin 30^\circ = 1/2$, the downward push increases N most, and the upward pull decreases N most. Thus, for $F > 0$:

$$N_d > N_c > N_a > N_b.$$

The question guarantees contact in a and b. At $F = 0$, all four normal forces would be equal.

Unit 2 rules to remember

1. Net force determines acceleration, not velocity.
2. Add force components separately along each axis.
3. Velocity-time slopes give acceleration components.
4. Normal force adjusts to maintain contact; it is not always mg .